

HR ANALYTICS — EMPLOYEE ATTRITION PREDICTION

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1. Introduction

Employee attrition is one of the most critical challenges faced by organizations today. When talented employees leave, companies incur significant costs related to recruitment, onboarding, and lost productivity. The HR Analytics — Employee Attrition Prediction project aims to address this challenge by leveraging data analytics and machine learning techniques to identify the key drivers of employee resignation and predict which employees are at the highest risk of leaving. This project was developed as part of the Elevate Labs Data Analyst Internship by Mohammad Sadique, a final-year BCA (Information Technology) student.

2. Abstract

This project analyzes the IBM HR Analytics Employee Attrition dataset containing 1,470 employee records with 35 features including age, department, salary, job satisfaction, overtime status, and years at the company. The analysis follows a structured pipeline: data cleaning and exploratory data analysis (EDA) using Python (Pandas and Seaborn), followed by building a Decision Tree classification model using Scikit-learn to predict attrition (Yes/No). SHAP-equivalent feature importance analysis identified Monthly Income, OverTime, and Age as the top three drivers of employee attrition. The model achieved approximately 85% accuracy. Key insights and risk factors were visualized in an interactive three-page Power BI dashboard using a professional Teal and Orange theme, enabling HR teams to take data-driven preventive action against employee turnover.

3. Tools Used

Tool	Purpose
Python (Pandas, NumPy)	Data loading, cleaning, and preprocessing
Seaborn & Matplotlib	Exploratory data analysis and visualizations
Scikit-learn (Sklearn)	Decision Tree model, train/test split, evaluation
Google Colab	Cloud-based Jupyter Notebook environment
Power BI Desktop	Interactive 3-page attrition dashboard
IBM HR Dataset (Kaggle)	1,470 employee records with 35 features

4. Steps Involved in Building the Project

- **Step 1 — Data Loading & Exploration:** Loaded the IBM HR Analytics CSV dataset using Pandas. Inspected shape (1,470 rows x 35 columns), data types, null values, and duplicate records. The dataset was confirmed clean with zero null values and no duplicates.
- **Step 2 — Exploratory Data Analysis (EDA):** Performed visual analysis using Seaborn and Matplotlib. Key charts included: overall attrition count (237 employees left = 16.12%), attrition by department (Sales highest), attrition by age (25-35 age group most vulnerable), monthly income vs attrition (lower income = higher attrition), and a correlation heatmap to identify feature relationships.
- **Step 3 — Data Preprocessing:** Dropped four redundant columns (EmployeeCount, Over18, StandardHours, EmployeeNumber). Applied LabelEncoder from Scikit-learn to convert all categorical text columns (Department, Gender, OverTime, etc.) into numeric format required by the ML model. Defined feature matrix X (30 columns) and target variable y (Attrition).

- **Step 4 — Train/Test Split:** Split the dataset into 80% training (1,176 rows) and 20% testing (294 rows) using `train_test_split` with `random_state=42` for reproducibility.
- **Step 5 — Model Training & Evaluation:** Trained a Decision Tree Classifier (`max_depth=5`) on the training set. Evaluated performance using accuracy score (~85%), confusion matrix, and classification report (Precision, Recall, F1-Score). Feature importance analysis revealed Monthly Income, OverTime, and HourlyRate as the top attrition predictors.
- **Step 6 — Power BI Dashboard:** Exported the cleaned dataset as CSV and built a 3-page interactive Power BI dashboard with Teal & Orange theme. Page 1: Executive Overview (KPI cards, donut chart, department/job role analysis). Page 2: Deep Dive Analysis (income, overtime, marital status, years at company). Page 3: Prevention Insights (top risk factors, high-risk employee table, job satisfaction analysis).

5. Conclusion

This project successfully demonstrated the application of data analytics and machine learning to solve a real-world HR business problem. The Decision Tree model achieved approximately 85% accuracy in predicting employee attrition. The analysis revealed that Monthly Income is the single most important factor — employees earning below average salary are significantly more likely to leave. OverTime emerged as the second most critical factor, with employees working overtime leaving at a much higher rate (416 cases) compared to those who do not (1,054 cases of no attrition). Research & Development recorded the highest number of departures (961), followed by Sales (446) and Human Resources (63). Married employees accounted for the highest attrition count (673), ahead of Single (470) and Divorced (327) employees, while attrition spiked sharply at the 5-year and 9-year tenure marks.

Key recommendations for HR teams: (1) Review compensation structures for low-income bands, (2) Monitor and limit mandatory overtime, especially in Research & Development and Sales, (3) Implement structured retention check-ins around the 5-year and 9-year tenure milestones, and (4) Conduct regular job satisfaction surveys, with focused attention on married employees who showed the highest attrition counts. The interactive Power BI dashboard provides HR managers with a real-time tool to monitor attrition risk and take timely preventive action.